From AlexNet to a Fully Convolutional AlexNet: Fewer Parameters, Better Accuracy on TinyImageNet

Nithin D H
Indian Institute of Technology Kanpur (IIT-Kanpur)
Kanpur, India

Abstract

AlexNet's three fully connected layers hold 94% of its parameters, yet they discard the spatial layout that the convolutional layers worked to build. In this project we convert AlexNet into a fully convolutional network (FCN). The backbone is kept exactly as published, and the fully connected head is replaced with 1×1 *convolutions and global average pooling.*

We train both models from scratch for 30 epochs on TinyImageNet (200 classes, 100k images), using the same data pipeline and optimiser. The FCN reaches ***37.0% top-1 / 63.2% top-5*** *validation accuracy, against 34.7% / 60.2% for the original CNN, while using* ***62% fewer parameters*** *(22.4 M vs. 59.1 M). Because it never flattens its features, the FCN also accepts larger test images without modification. We explain where the parameter savings come from, how the converted head differs from the classic weight-equivalent conversion, and what the training curves do and do not tell us about why the FCN generalises better.*

1. Introduction

The flatten bottleneck. In AlexNet [1], the $6 \times 6 \times 256$ output of the last pooling layer is flattened into a 9,216-dimensional vector and passed to fully connected layers. The first of these, FC6, alone has 37.8 M weights. Flattening fixes the input resolution, because any other image size changes the vector length, and it ties each weight to one absolute spatial position.

Our approach. We keep the convolutional backbone unchanged and rewrite the head as convolutions: FC6, FC7 and FC8 become 1×1 convolutions that score every cell of the 6×6 grid, and global average pooling (GAP) collapses the resulting score map into 200 logits. To leave the backbone untouched, the 64×64 TinyImageNet images [2] are upsampled to the original 227×227 input size.

2. Related Work

Convolutionalising classifiers. OverFeat [5] and FCN [4] observed that a fully connected layer applied to a $k \times k$ feature map equals a $k \times k$ convolution. Converting FC6 to a 6×6 convolution and FC7/FC8 to 1×1 convolutions lets a classifier slide over larger images and produce dense score maps, which is the basis of FCN segmentation.

Global average pooling heads. Network-in-Network [3] and GoogLeNet [7] replaced large fully connected heads with per-location classifiers followed by GAP, cutting parameters and acting as a structural regulariser. Our head follows this design.

3. Method

Figure 1 shows both architectures and Table 1 the parameter breakdown.

3.1. Shared backbone

Both models use the original AlexNet feature extractor: Conv1 (96@11×11, stride 4) with local response normalisation (LRN), max-pooling, Conv2 (256@5×5) with LRN, max-pooling, Conv3–Conv5 (3×3 with 384, 384 and 256 filters), and a final max-pooling. For a 227×227 input this produces a $6 \times 6 \times 256$ map; the backbone has 3.75 M parameters.

3.2. The CNN head

The map is flattened to 9,216 values and passed through FC6 (9216→4096), FC7 (4096→4096) and FC8 (4096→200), with ReLU and dropout ($p = 0.5$) before FC6 and FC7.

3.3. The FCN head

Nothing is flattened. Conv-FC6 is a 1×1 convolution from 256 to 4,096 channels, Conv-FC7 a 1×1 convolution from 4,096 to 4,096, and Conv-score a 1×1 convolution to 200 channels. This yields a $6 \times 6 \times 200$ class-score map, one prediction per spatial cell, and GAP averages the 36 predictions into the final logits. Dropout2d ($p = 0.5$) precedes Conv-FC6 and Conv-FC7.

Where the savings come from. FC6 connects every one of the 9,216 flattened values to 4,096 units (37.75 M weights). Conv-FC6 connects only the 256 channels at one location and shares those weights across all 36 locations (1.05 M weights, 36× fewer). FC7 and FC8 are unchanged in size.

Not a weight-for-weight equivalent. The exact convolutional rewrite of FC6 is a 6×6 convolution with the same

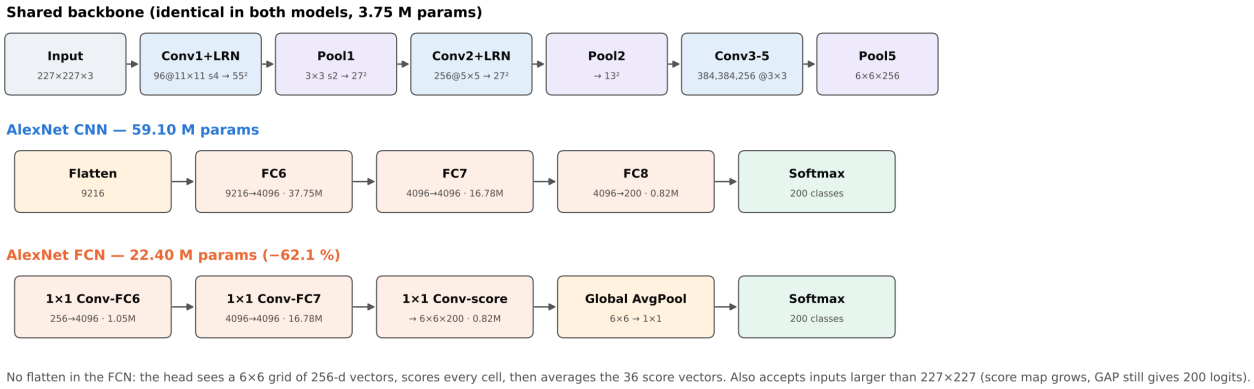


Figure 1. Architectures. Top: shared AlexNet backbone (identical in both models). Middle: the original fully connected head. Bottom: the fully convolutional head. Parameter counts are for a 200-class output.

Table 1. Layer-by-layer parameters (verified by instantiating both models).

Layer	AlexNet CNN	AlexNet FCN
Backbone (Conv1–5)	3,747,200	3,747,200
FC6 / Conv-FC6	37,752,832	1,052,672
FC7 / Conv-FC7	16,781,312	16,781,312
FC8 / Conv-score	819,400	819,400
Total	**59,100,744**	**22,400,584**

37.75 M weights [4]; it computes the identical function and leaves the parameter count unchanged. Our 1×1 + GAP head is a different, lighter model: it gives up absolute position information in exchange for weight sharing across positions. The comparison below is therefore between two architectures, not two parameterisations of one function.

4. Results and Discussion

4.1. The data

TinyImageNet [2] contains 200 ImageNet classes with 500 training and 50 validation images each, at 64×64 pixels. Images are resized to 256 and cropped to 227 (random crop for training, centre crop for validation), with horizontal flips and colour jitter (brightness, contrast and saturation 0.3, hue 0.1) during training.

4.2. Implementation details

Both models are trained from scratch for 30 epochs with SGD (momentum 0.9, weight decay 10^{-4}), batch size 128, gradient clipping at norm 2.0 and a one-cycle learning-rate schedule [6] with peak 10^{-3}. Weights are initialised from $\mathcal{N}(0, 0.01)$, with AlexNet's bias of 1 on FC6 and FC7 of the CNN. Training ran on a single NVIDIA T4 GPU. In the run reported here the one-cycle scheduler was stepped once per epoch; the released code steps it once per batch, as intended, so re-running it will likely change the absolute accuracies of both models.

Table 2. Best validation metrics over 30 epochs (values marked ∼ are read off the curves in Fig. 2).

Metric	CNN	FCN	Δ
Top-1 accuracy	34.7%	**37.0%**	+2.3
Top-5 accuracy	60.2%	**63.2%**	+3.0
Final validation loss	∼2.77	**∼2.60**	−0.17
Final train top-1	∼40%	∼43%	+3
Epoch time (T4)	∼140 s	∼148 s	+8 s
Parameters	59.10 M	**22.40 M**	−62%

4.3. Quantitative results

Table 2 summarises the run and Figure 2 shows the full training curves.

4.4. Qualitative analysis

Convergence. Both models train smoothly. The FCN is ahead on the validation curves from the first epochs and on training loss from about epoch 8, and the gap widens slowly up to epoch 30. Validation accuracy is still rising at the end, so neither model has converged.

Why might the FCN generalise better? The train–validation top-1 gap is similar for the two models (∼5–6 points), so the CNN is not simply overfitting more. Two explanations fit the curves: the averaged per-cell predictions act like a built-in ensemble over 36 shifted views, and the much smaller head is easier to optimise from scratch. This run cannot separate the two; an ablation with a 6×6-convolution head would isolate the effect of weight sharing.

Fewer parameters is not faster. Despite 62% fewer weights, the FCN is ∼6% slower per epoch. Compute is dominated by the backbone, and Conv-FC7 now runs at 36 locations instead of one.

Variable input size. Because the head never flattens, the FCN accepts larger inputs at test time: a 320×320 image yields a larger score map that GAP still reduces to 200 logits.

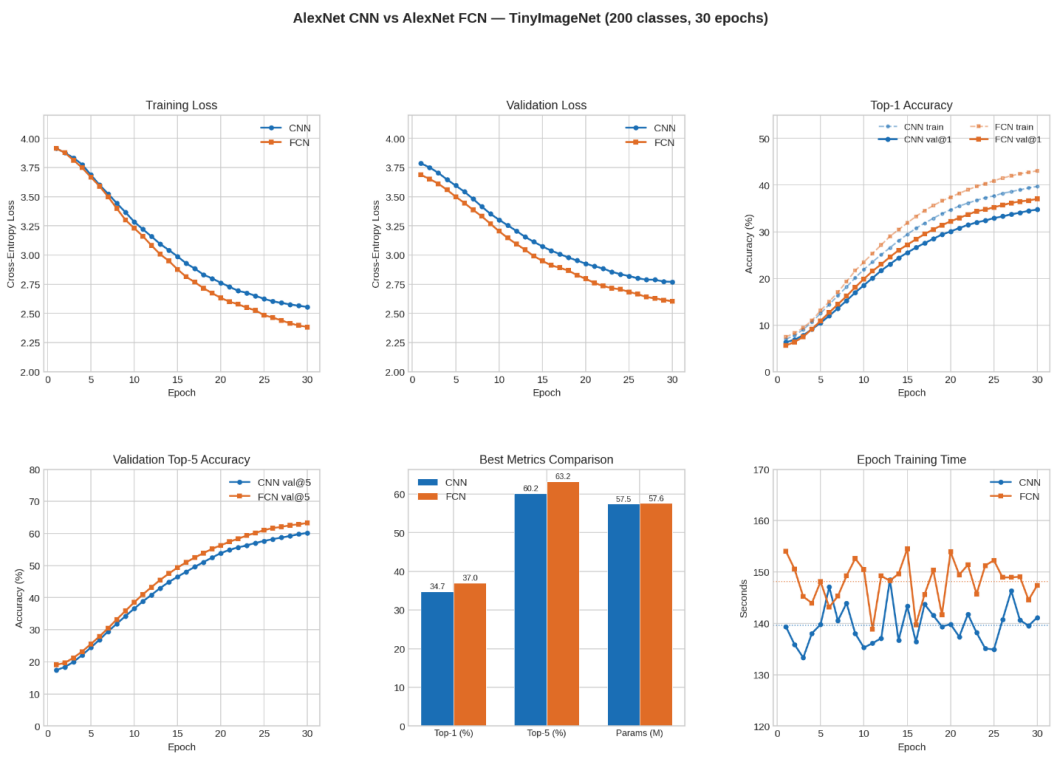


Figure 2. Training and validation loss, top-1 and top-5 accuracy, best-metric comparison and epoch time for both models over 30 epochs.

The CNN only accepts 227×227.

5. Conclusion & Future Work

Replacing AlexNet's fully connected head with 1×1 convolutions and global average pooling cut the parameter count by 62% and *improved* TinyImageNet accuracy by 2.3 (top-1) and 3.0 (top-5) points under an identical training setup. The FCN also gains resolution independence.

Limitations. Each model was trained once, so a 2–3 point difference should be confirmed over several seeds. Upsampling 64×64 images to 227×227 adds compute but no information. AlexNet is a 2012 baseline; the focus here is the head design, not state-of-the-art accuracy.

Next steps. (i) Re-run with per-batch scheduling and longer training; (ii) add the weight-equivalent 6×6 head as a third arm; (iii) visualise the 6×6 score maps as class activation maps, which the FCN provides at no extra cost; and (iv) evaluate at multiple test resolutions.

Acknowledgement. This work was carried out as part of EE655 Computer Vision & Deep Learning at IIT Kanpur under the supervision of Dr. Koteswar Rao Jerripothula.

References

[1] A. Krizhevsky, I. Sutskever, and G. E. Hinton. ImageNet classification with deep convolutional neural networks. In *Adv. Neural Inform. Process. Syst.*, 2012.

[2] Y. Le and X. Yang. Tiny ImageNet visual recognition challenge. CS231N, Stanford University, 2015.

[3] M. Lin, Q. Chen, and S. Yan. Network in network. In *Int. Conf. Learn. Represent.*, 2014.

[4] J. Long, E. Shelhamer, and T. Darrell. Fully convolutional networks for semantic segmentation. In *IEEE Conf. Comput. Vis. Pattern Recog.*, 2015.

[5] P. Sermanet, D. Eigen, X. Zhang, M. Mathieu, R. Fergus, and Y. LeCun. OverFeat: Integrated recognition, localization and detection using convolutional networks. In *Int. Conf. Learn. Represent.*, 2014.

[6] L. N. Smith and N. Topin. Super-convergence: Very fast training of neural networks using large learning rates. In *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications*, SPIE, 2019.

[7] C. Szegedy et al. Going deeper with convolutions. In *IEEE Conf. Comput. Vis. Pattern Recog.*, 2015.